Picard Iteration: For an IVP with

$$x'(t) = v(x(t), t), \quad x(t_0) = x_0$$

define

$$X_0 = x_0,\ X_1 = x_0 + \int_{t_0}^{t} v(x_0(s), s)\,\mathrm{d}s,\ \ldots,\ X_n = x_0 + \int_{t_0}^{t} v(x_{n-1}(s), s)\,\mathrm{d}s.$$

The solution to the IVP is then $\lim_{n\to\infty} X_n$

Supremum Distance: For f, g continuous functions on $[a, b]$ define

$$d(f, g) = \max_{t\in[a,b]} |f(t) - g(t)|.$$

Tip: Square the distance to remove the absolute value bars

Second Order ODEs: For an equation in the form

$$ax''(t) + bx'(t) + cx(t) = f(t)$$

start by solving the homogenous equation where $f(t) = 0$ and $a, b, c \in \mathbb{R}$, with the characteristic equation $ar^2 + br + c = 0$.

- If $r_1 \neq r_2$ then $x_h = C_1e^{r_1 t} + C_2e^{r_2 t}$
- If $r_1 = r_2$ then $x_h = (C_1 + C_2t)e^{r_1 t}$

Then solve for the particular solution by guessing:

- $g(x) = e^{\alpha t} \Rightarrow x_p = Ce^{\alpha t}$
- $g(x)$ is an n-degree polynomial $\Rightarrow x_p = A_nt^n + \cdots + A_1t + A_0$ (can ommit constant term based on context)
- $g(x) = A\cos bt$ or $A\sin bt \Rightarrow x_p = C\cos bt + D\sin bt$.

If the x_p guess is already accounted for in the homogenous solution, multiply by t until it is unique. The final solution is then $x = x_p + x_h$.

Cauchy-Euler Equations: For an equation in the form

$$t^2x'' + atx' + bx = g(t)$$

guess $x_1 = t^\alpha$ so that $x_1' = \alpha t^{\alpha-1}$ and $x_1'' = \alpha(\alpha - 1)t^{\alpha-2}$.

- If $\alpha_1 \neq \alpha_2$ then $x_h = C_1t^{\alpha_1} + C_2t^{\alpha_2}$
- If $\alpha_1 = \alpha_2$ then $x_h = C_1t^{\alpha_1} + C_2t^{\alpha_1}\ln t$

Then find the particular solution by guessing or variation of parameters. The final solution is then $x = x_1 + x_p$.

Variation of Parameters: If $x_1(t)$ and $x_2(t)$ are solutions to the homogenous equation then a particular solution is

$$x_p = \int_{t_0}^{t} = \frac{\det(N(t, s))}{\det(M(s))} f(s)\,\mathrm{d}s$$

for

$$N(t, s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1(t) & x_2(t) \end{pmatrix} \quad \text{and} \quad M(s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1'(s) & x_2'(s) \end{pmatrix}$$

Solution Space Dim: A nth order differential equation has a solution space with order n. Thus, any solution set with more than n solutions, must contain at least one linearly dependent solution.

Good to know formulas:

$$e^t = \sum_{k=0}^{\infty} \frac{t^k}{k!} = 1 + t + \frac{t^2}{2!} + \frac{t^2}{3!} + \cdots$$

$$te^t = \sum_{k=0}^{\infty} \frac{t^{k+1}}{k!} = t + t^2 + \frac{t^3}{2!} + \frac{t^3}{3!} + \cdots$$

$$\sin t = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} t^{2k+1} = t - \frac{t^3}{3!} + \frac{t^5}{5!} + \cdots$$

$$\cos t = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+)!} t^{2k} = 1 - \frac{t^3}{3!} + \frac{t^4}{4!} + \cdots$$

$$\sin 2t = 2\sin t\cos t$$

$$\cos 2t = \cos^2 t - \sin^2 t$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

$$\frac{1}{1-t} = \sum_{t=0}^{\infty} t^n = 1 + t + t^2 + \cdots$$